Fleet Idling and Alerts Analytics: Full Build Documentation

This document records everything built in the dashboard: the data pipeline, the model, every page (sheet) with its cards, visuals, fields and filters, the measures and columns, the problems met on the way and how they were fixed, and how to maintain it.

1. Summary

- **Goal:** show where, when and how much vehicles idle, what it costs in fuel, and how much is avoidable.
- **Period:** 1 Jan 2026 to yesterday, refreshed daily. **Fleets:** 18. **Vehicles:** 1,966 in the Fleet Master, 1,822 active.
- **Pages:** 8 report pages, plus a front page, a navigation cover, an ERD board and a print page (design boards).
- **Stack:** Intangles API, Python, SQL Server Express (db_idling), Windows Task Scheduler, Power BI Desktop.

2. Data pipeline

1. **Extract:** `idling_etl.py` calls the Intangles alert-logs API for every active fleet (account id, start and end time in ms, type idling, paged). It uses 3-day windows, a 0.5 s pause between calls and 6 retries with growing waits for HTTP 429 and 5xx.
2. **Bronze:** raw rows (plus the raw JSON) are appended to `bronze_idling_alerts` with a batch id.
3. **Silver:** `usp_bronze_to_silver` merges on `alert_id` (no duplicates), converts UTC to IST and adds date, hour, minutes and hub fields.
4. **Gold:** `usp_build_gold` rebuilds daily, vehicle, hub and hourly summaries. The Power BI model reads silver, not gold.
5. **Incremental load:** `etl_watermark` stores the last loaded time per fleet. Each run re-reads the last 24 hours so late alerts are not missed. Progress is saved after every window, so an interrupted run resumes.
6. **Backfill:** `--from-date` and `--to-date` load a fixed history range without touching the watermark. History from 1 Jan 2026 was loaded this way.
7. **Schedule:** Task Scheduler runs `cmd.exe /c run_idling_etl.bat` daily at 10:00 AM IST and writes `etl_run_log.txt`. Status per fleet is in `etl_log` (OK, WARN_ZERO_ROWS or ERROR).

8. **Vehicle master:** `load_vehicle_master.py` cleans the Fleet Master Excel and loads `dim_vehicle` (see section 4).
9. **Planned:** Power BI Service with an on-premises gateway and a scheduled refresh about 30 minutes after the ETL.

3. Data model

Table	Role	Key columns
silver_idling_alerts	Fact, one row per alert	alert_id, account_id, vehicle_id, alert_date, alert_hour, duration_min, duration_sec, idle_threshold_sec, fuel_litres, hub_name, hub_distance_m, inside_hub, latitude, longitude, address
dim_date	Calendar from 2026	date_key, day_name, month_name, month_no, year_no
dim_fleet	Fleet list	account_id, fleet_name, is_active
dim_vehicle	Cleaned fleet master	vehicle_id, master_account_id, plate, vehicle_type, ac_type, ac_detail, make, engine_type, powertrain, operational_type, vehicle_status, is_active, fuel_data, camera, data_type
Fuel Parameter	What-if slider, fuel price	Fuel Parameter, Fuel Parameter Value
Reduction Target	What-if slider, % cut	Reduction Target, Reduction Target Value

Relationships (one to many, single direction): dim_fleet[account_id] to silver[account_id]; dim_date[date_key] to silver[alert_date]; dim_vehicle[vehicle_id] to silver[vehicle_id].

dim_fleet and dim_vehicle are deliberately not related. Active Vehicles links them in DAX with TREATAS (account_id to master_account_id). Any relationship Power BI creates on master_account_id or plate must be deleted.

4. Vehicle master cleaning rules

- **AC type:** AC, 9M AC, 12M AC and Sleeper AC become AC; NAC becomes Non-AC; Not Available, Sold and blank become Unknown. `ac_detail` keeps 9 m, 12 m and Sleeper.

- **Engine type:** BS3, BS4, BS6 kept; Electric buses become Electric; 0, Sold and blank become Unknown.
- **Make:** Daimler and BharatBenz merged as BharatBenz (Daimler); Toyota and Toyota Kirloskar merged; names tidied.
- **Vehicle type:** Car, Innova and Etios grouped as Car / SUV; Sold becomes Unknown.
- **Powertrain (derived):** Electric for Electric Bus, CNG for the Chartered CNG fleet, otherwise Diesel / Other ICE.
- **is_active:** Operational and not sold. Result: 1,822 active of 1,966.
- **Join key:** the master `id` equals the alert `vehicle_id` (plates are not used because they vary).

5. Common page setup

- Theme `Idling_Theme.json` (amber and navy), canvas 16:9.
- Navy header band with the page title; Date slicer (dim_date[date_key], between), Fleet slicer (dropdown) and Fuel Parameter slicer, synced across pages.
- Row of six cards, then two rows of visuals, then a `Data Updated (IST)` card at the bottom.
- Titles 13 pt bold navy, subtitles 10 pt grey. Red means worse, green means better.
- Trend charts leave out today (Relative date filter, last 365 days, Include today off) because today is partial.

6. Page by page

Page 1: Executive Summary

Question: how big is the problem and is it improving?

Item	Type	Fields
Cards	6 cards	Total Alerts, Idle Hours, Excess Idle Hours, Idle Cost Display, Avoidable Idle Cost (Rs), Alerts per Active Vehicle
Daily Idling Alerts Trend	Line	date_key; Total Alerts and Alerts 7D Avg
Idle Hours by Fleet	Bar, top 10	fleet_name; Idle Hours
Auto insights	3 cards	Insight - Trend, Insight - Worst Fleet, Insight - Peak Hour

Item	Type	Fields
Monthly Idle Cost	Column	Month Year; Idle Cost (Rs); tooltip Idle Cost MoM %
Monthly Alerts and MoM %	Column and line	Month Year; Total Alerts; Alerts MoM %

Notes: Insight - Trend compares the chosen range with the month before, so pick one full month for a meaningful sentence.

Page 2: Fleet Scorecard

Question: which fleets are worst, compared fairly?

Item	Type	Fields
Cards	6	Active Vehicles, Active Vehicles Affected, Vehicles with Idling %, Idle Hours per Vehicle, Avg Severity (x), Idle Change 30D %
Fleet Scorecard	Table	fleet_name, Active Vehicles, Active Vehicles Affected, Vehicles with Idling %, Total Alerts, Idle Hours, Idle Hours per Vehicle, Excess Idle Hours, Avg Severity (x), Idle Change 30D %
Idle Hours per Vehicle	Bar, top 10	fleet_name; Idle Hours per Vehicle
Fleet Positioning	Scatter	X Alerts per Active Vehicle; Y Avg Severity (x); size Idle Hours

Formatting: data bars on idle hours, red font for rising and green for falling on Idle Change 30D %. Per-vehicle measures make large and small fleets comparable.

Page 3: Vehicle Analysis

Question: do AC, engine type, make or vehicle type change idling?

- **Slicers:** vehicle_type, ac_type, make, engine_type, powertrain, operational_type.
- **Cards:** Active Vehicles, Vehicles with Idling %, Idle Hours per Vehicle, Alerts per Active Vehicle, Avg Severity (x), Fuel per Idle Hour.
- **Visuals:** AC vs Non-AC (bar), by engine type (bar), by make (bar), engine type by AC matrix, by vehicle type (column). All use Idle Hours per Vehicle or Alerts per Active Vehicle.

Page 4: Top Vehicles

Question: which vehicles idle the most, and most often?

Item	Type	Fields
Cards	6	Top 10 Idle Hours, Top10 Share %, Repeat Offender Vehicles, Longest Idle (min), Avg Severity (x), Active Vehicles Affected
Top 15 Vehicles by Idle Hours	Bar	vehicle_plate (Top N 15 by Idle Hours); Idle Hours; legend fleet_name
Repeat Offenders	Table	vehicle_plate, fleet_name, vehicle_type, ac_type, engine_type, Idle Days, Idle Hours, Avg Severity (x), Last Alert (IST); filter Idle Days at least 100
Chronic vs Severe Idlers	Scatter	details vehicle_plate; X Idle Days; Y Avg Severity (x); size Idle Hours
Idle Length Buckets	Column	Idle Bucket (sorted by Bucket Sort); Total Alerts

A vehicle_plate dropdown slicer sits in the header. A low Top10 Share (about 6%) shows idling is a fleet-wide habit, not a few vehicles.

Page 5: Time Patterns

Question: when does idling happen?

Item	Type	Fields
Cards	6	Night Alerts %, Weekend Share %, Avg Alerts per Day, Alerts 30m+, Spike Days, Long Idle %
Alerts by Hour of Day	Column	alert_hour (categorical axis); Total Alerts
Alerts by Shift	Column	Shift (sorted by Shift Sort); Total Alerts
Shift Share	Donut	Shift; Shift Share %
Weekday x Hour Heatmap	Matrix	day_name (sorted by Day No) by alert_hour; Total Alerts; colour scale
Weekday vs Weekend	Column	Day Category; Avg Alerts per Day
Peak time	Card	Insight - Peak Hour

Weekday and weekend are compared on average per day because there are five weekdays and two weekend days. Shifts: Night 12am-6am, Morning 6am-12pm, Afternoon 12pm-6pm, Evening 6pm-12am. Night Alerts % uses 10pm to 5am.

Page 6: Hub and Location

Question: inside or outside depots, and where?

Item	Type	Fields
Cards	6	Outside Hub %, Inside Hub Idle %, Outside Hub Idle Hours, Long Idle Alerts (2h+), Total Alerts, Idle Hours
Hub Status	Donut	Hub Status; Total Alerts
Top 10 Hubs	Bar	Hub Clean (top 10, No hub data left out); Total Alerts
Distance from Hub	Column	Hub Distance Bucket; Total Alerts
Hub Status by Fleet	100% stacked bar	fleet_name; Hub Status
Repeat Locations	Table	Location Key (top 20), Total Alerts, Idle Hours, Location Link (web URL)

Some fleets send no hub data and show as No hub data. The map visual is blocked by the tenant admin, so a table with a Google Maps link is used.

Page 7: Insights and Savings

Question: what is avoidable and what would a reduction save?

Item	Type	Fields
Cards	6	Avoidable Idle Cost (Rs), Savings at Target (Rs), Avg Severity (x), Spike Days, Repeat Offender Vehicles, Idle Change 30D %
Sliders	What-if	Fuel Parameter, Reduction Target
Idling Severity	Column	Severity Band; Idle Hours
Last 30 vs Previous 30 Days	Table	fleet_name, Idle Hours Last 30D, Idle Hours Prev 30D, Idle Change 30D % (red up, green down)

Item	Type	Fields
Long Idling Trend	Line	date_key; Alerts 30m+ and Excess Idle Hours
What Drives Idling	Decomposition tree	analyze Idle Hours; explain by fleet_name, Hub Clean, Shift, Day Category, vehicle_plate
What Makes Idling Long	Key influencers	analyze Is Long Idle; explain by shift, hub status, day type, vehicle attributes
Summary	Smart narrative	automatic text

Savings at Target = Avoidable Idle Cost x Reduction Target / 100, so 100% equals the avoidable cost. Key influencers must not include fleet_name for the first reading, because thresholds differ by fleet and make small fleets look extreme.

Page 8: Alert Detail (Exceptions)

Question: which are the longest idling events?

- **Page filter:** duration_min at least 120.
- **Cards:** Total Alerts (long idle alerts), Longest Idle (min), Avg Idle (min), Excess Idle Hours, Fuel Wasted (L), Active Vehicles Affected.
- **Long Idling Events:** table with time (IST), fleet, vehicle, address, idle minutes, fuel litres, hub and a Maps link, longest first, totals off.
- **Long Idles by Fleet:** bar, top 10.
- A text box explains how to read the page. Drill-through on vehicle_plate opens this page from other pages.

Design boards

- **Front page:** A detailed overview (data scope, current thresholds, key and operational insights, core metrics) in the style of the Qlik example, with room for the bus photo.
- **00 Cover:** a navigation cover with eight tiles that link to the pages.
- **ERD:** star schema with the data flow and control tables.
- **Print page:** A4 landscape version with the bus photo, for printing.

7. Idling thresholds

Threshold	Fleets (current)
5 min	Chartered CNG, Staff Operations
10 min	Chartered Arunachal EV, Chartered INTAS, Chartered Speed-OSRTC, GIFT GO, Indore Intracity
15 min	AMNS Hazira, DCM Diesel, DCM SRIRAM, Guwahati Intercity, Indore Electric, Rajasthan EV, School Operations
20 min	Ahmedabad GSRTC, Ahmedabad Intercity, Indore Intercity
30 min	Chartered Speed Ltd.

Thresholds were raised during 2026 for most fleets (some fleets had two or three values), so alert counts fell without a change in behaviour. For trends use Excess Idle Hours and Alerts 30m+.

8. Calculated columns

Table	Columns
silver_idling_alerts	Hub Clean, Hub Status, Idle Bucket, Bucket Sort, Shift, Shift Sort, Day Category, Night Idling, Hub Distance Bucket, Location Key, Location Link, Maps Link, Threshold Minutes, Idle Multiple, Severity Band, Is Long Idle
dim_date	Day No, Month Sort, Month Year

Idle Multiple is duration divided by threshold. Severity Band: Mild under 1.5x, Moderate 1.5 to 3x, High 3 to 6x, Severe 6x and above.

9. Measures

Measure	Definition
Total Alerts	Count of alert rows

Measure	Definition
Idle Hours	Sum of duration_min divided by 60
Excess Idle Hours	Sum of (duration minus threshold, not below 0) divided by 3600
Fuel Wasted (L)	Sum of fuel_litres, electric vehicles excluded
Idle Cost (Rs)	Fuel Wasted (L) x Fuel Parameter Value
Idle Cost Display	Idle Cost as text in Rs, Lakh or Cr (cards only)
Avoidable Idle Cost (Rs)	Idle Cost x Excess Idle Hours / Idle Hours
Savings at Target (Rs)	Avoidable Idle Cost x Reduction Target Value / 100
Active Vehicles	Distinct active vehicles in dim_vehicle for the chosen fleets (TREATAS)
Active Vehicles Affected	Distinct vehicle_id with alerts among active vehicles
Vehicles with Idling %	Active Vehicles Affected / Active Vehicles
Idle Hours per Vehicle, Alerts per Active Vehicle	Idle Hours or Total Alerts divided by Active Vehicles
Avg Severity (x)	Average of Idle Multiple (format as a number, not a percentage)
Idle Days	Distinct alert dates
Repeat Offender Vehicles	Vehicles with Idle Days of 100 or more
Top 10 Idle Hours, Top10 Share %	Idle hours of the top 10 vehicles, and their share of all idle hours
Alerts MoM %, Idle Cost MoM %	Change against the previous month (DATEADD on dim_date)
Idle Hours Last 30D, Prev 30D, Idle Change 30D %	Rolling windows ending at the latest alert date, ignoring the date slicer
Alerts 7D Avg	7-day average of Total Alerts
Alerts 30m+	Alerts of 30 minutes or more
Spike Days	Days with alerts above twice the daily average

Measure	Definition
Night Alerts %, Weekend Share %, Shift Share %	Shares of Total Alerts
Avg Alerts per Day	Total Alerts / distinct alert dates
Long Idle Alerts (2h+), Long Idle %	Alerts of 120 minutes or more, and their share
Inside Hub Idle %, Outside Hub %	Idle hour shares by Hub Status, ignoring No hub data
Insight - Trend, Worst Fleet, Peak Hour	Sentences built from the measures above
Data Updated (IST)	Latest load time plus 5 hours 30 minutes

10. Problems met and fixes

Problem	Cause	Fix
Retries exhausted in ETL	API rate limit (HTTP 429)	Throttle, 6 retries with growing waits
Task Scheduler error 0x2	Path with spaces	Run through cmd.exe /c with Start in set
Late alerts missed	Alert arrives after idling ends	24-hour overlap, silver MERGE removes duplicates
Single value cannot be determined	A row-by-row formula built as a measure	Create it as a New column
PLACEHOLDER error	Filtering on a broken measure	Make the field a real column first
Column name conflict	Old measure with the same name	Delete the old item or use a new name (Day Category)
Date slicer not filtering dim_date	Slicer built on silver alert_date	Use dim_date[date_key] and filter by Total Alerts above 0
Only one Top N filter allowed	Copied visual still had Top N on another field	Remove the old field, then set Top N
Avg Severity shown as 214%	Percentage format	Decimal number, 2 places
Dollar sign on costs	Wrong currency format	Currency, Rs English (India)

Problem	Cause	Fix
Fuel for electric vehicles	EV fuel values are not reliable	Exclude powertrain Electric from fuel and cost
Savings above avoidable cost	Savings used total cost	Savings now uses Avoidable Idle Cost
Jan drop in alerts	Thresholds raised	Compare with Excess Idle Hours
Cost differed between pages	Fuel slider not synced	Sync the Fuel Parameter slicer

11. Validation checklist

- Total Alerts about 366K and Idle Hours about 95K for the full period.
- Active Vehicles equals 1,822; Vehicles with Idling % stays under 100%.
- Page 6 Long Idle Alerts (2h+) equals Total Alerts on page 8.
- Savings at Target equals Avoidable Idle Cost when the slider is 100.
- Selecting one fleet changes every card and chart on the page.
- Compare monthly idle hours with the old Qlik dashboard; small gaps are expected because only idling above the threshold is recorded.

12. Maintenance

- **Daily:** the ETL runs by itself; press Refresh in Power BI until the gateway is set up.
- **When the vehicle master changes:** update the Excel and run `python load_vehicle_master.py`, then refresh.
- **New fleet:** insert a row in dim_fleet.
- **Token:** rotate the Intangles token and run `setx INTANGLES_TOKEN` with the new value.
- **Storage:** check `EXEC sp_spaceused` now and then (Express limit is 10 GB per database); take a database backup regularly.
- **Sharing:** give others a read-only SQL login or a backup file; never share the token.

13. Known limits

- Only idling above the threshold is recorded.
- Alert counts are not comparable across fleets with different thresholds.
- Hub data is missing for some fleets; some vehicles have Unknown type, AC or make, and a few alerting vehicles are not in the master.

- Engine hours are not in the data, so Idling % of engine hours (shown in the old Qlik report) cannot be calculated.
- The Fleet Master is a snapshot dated 17 Oct 2026.
- Map visuals are disabled by the tenant admin; AI visuals may also be restricted.